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# ABSTRACT

Application Programming Interfaces (APIs) have become an important part of modern applications by enabling communication between different services. As the number of APIs continues to grow, they have become a common target for cyberattacks, making API security an important challenge. Existing API security solutions mainly rely on predefined rules and signature-based detection methods, which may not effectively identify new attack patterns or suspicious user behavior. This project proposes an Intelligent API Threat Detection framework using Graph Learning that represents API interactions as a graph of clients, users, endpoints and resources to capture the relationships between them. A Graph Attention Network (GAT) learns from this structure and from node and edge features to identify unusual API activity that may indicate a security threat. A Random Forest classifier on tabular request features serves as a baseline. Each analyzed request is assigned a threat class and a risk score between 0 and 100. Explainable Artificial Intelligence (XAI) is integrated to give clear and understandable explanations for detected anomalies, improving the transparency and interpretability of the detection process. The system is implemented as a FastAPI backend with a MySQL store and a web dashboard that shows threat overviews, risk-score distribution, threat-type distribution, the API interaction graph and model explanations. In the current prototype, the Random Forest baseline reaches an accuracy of 99.86% and the Graph Attention Network reaches 98.49%. The prototype graph contains 354 nodes and 10,866 edges. The project aims to provide an accurate, reliable and explainable approach to API threat detection while offering an efficient and scalable solution for modern API-based applications.